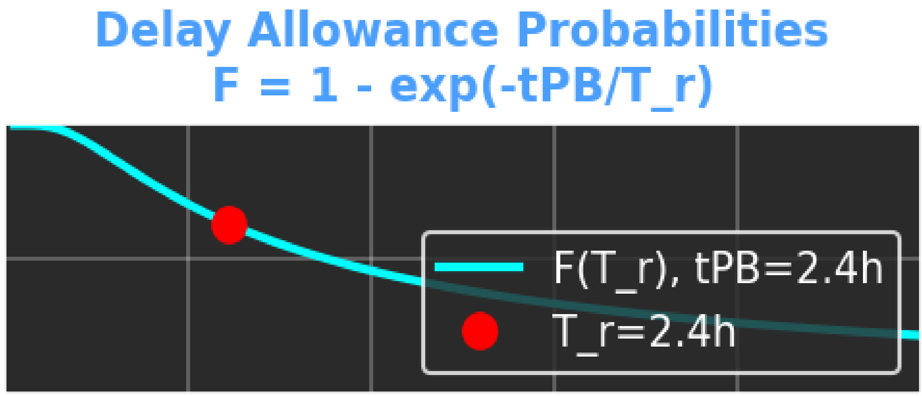
■ Power System Simulation Report

Simulation Description

This report contains the results of a Seven Bus Power System simulation with Battery Energy Storage Systems (BESS). The simulation analyzed fault conditions and battery compensation strategies.

Simulation Parameters:
• Recovery Time (T_r): 146.9 minutes
• Simulation Mode: manual
• Faulted Bus: Bus 1
• Fault Type: slg
• Timestamp: 2025-09-21 14:40:58

Delay Allowance Probabilities



Bus Voltage States Table

Bus	Pre-fault	During Fault	Post-fault	Compensated	While Charging	Normal
Bus 1	1.000∠0.0°	0.000∠0.0°	DISCONNECTED	1.000∠0.0°	1.000∠0.0°	1.000∠0.0°
Bus 2	0.937∠-4.4°	0.490∠-29.9°	0.835∠-12.5°	1.002∠-4.5°	0.927∠-9.2°	0.937∠-4.4°
Bus 3	0.920∠-5.5°	0.521∠-0.8°	0.827∠-13.2°	1.000∠-5.6°	0.908∠-11.1°	0.920∠-5.5°
Bus 4	0.930∠-4.7°	0.520∠-0.7°	0.837∠-12.3°	1.000∠-4.8°	0.918∠-10.1°	0.930∠-4.7°
Bus 5	0.927∠-4.8°	0.546∠-1.4°	0.843∠-11.9°	1.000∠-4.9°	0.914∠-10.6°	0.927∠-4.8°
Bus 6	0.940∠-4.0°	0.566∠-31.8°	0.865∠-10.5°	1.003∠-4.0°	0.929∠-9.6°	0.940∠-4.0°

Bus 7	1.000∠2.1°	0.760∠-1.4°	1.000∠0.0°	1.000∠2.0°	1.000∠-3.5°	1.000∠2.1°

Column Explanations

1. Pre-fault: Normal operating conditions before fault occurrence.
Equation: Standard power flow: $P + jQ = V * I^*$

2. During Fault: System conditions during fault event.
Equation: Fault current: $I_f = V_f / Z_f$

3. Post-fault: System state after fault clearance and isolation.
Equation: Modified network with faulted elements removed

4. Compensated: Voltages with battery compensation active.
Equation: $P_{bess} = (E*(SOC-SOC_{min})*\eta_{dis})/t_{discharge}$
$Q_{bess} = Kq*(V_{ref} - |V|)$

5. While Charging: System with increased load for battery charging.
Equation: $P_{load\_total} = P_{load\_original} + P_{charging}$

6. Normal: Restored normal operating conditions.
Equation: Standard power flow restoration

Key Equations

Battery Discharge Time:
$tPB = (E_{rated} * (SOC - SOC_{min}) * \eta_{discharge}) / P_{discharge}$

Probability Function:
$F(T_r) = 1 - \exp(-tPB/T_r)$

Droop Control:
$Q = Kq * (V_{ref} - |V|)$

SOC Update:
$SOC(t+\Delta t) = SOC(t) - (P_{discharge} * \Delta t) / (E_{rated} * \eta_{discharge})$